

DevSight AI: AI-Powered DevOps & Code Intelligence Platform

Section – 6

1. Abstract

Modern software projects often consist of large codebases, multiple frameworks, and complex architectures, making them difficult for developers to understand and maintain. New contributors spend significant time exploring project structures before contributing effectively, while ensuring code quality, detecting security vulnerabilities, and configuring deployment environments typically require multiple tools and considerable manual effort. This fragmented workflow reduces productivity, delays software delivery, and increases the risk of errors.

DevSight AI is an AI-powered DevOps and Code Intelligence Platform that streamlines software development by automatically analyzing GitHub repositories or project archives. It identifies programming languages, frameworks, dependencies, and project architecture while evaluating code quality and detecting security vulnerabilities. The platform also generates documentation, Dockerfiles, CI/CD workflows, and Kubernetes deployment manifests, and includes an AI-powered chatbot to answer repository-specific questions and explain source code. By integrating AI with DevOps and DevSecOps practices, DevSight AI reduces developer onboarding time, improves code quality, enhances application security, and accelerates software delivery through a single unified platform.

2. Problem Statement

Software development teams frequently work with large and complex codebases that are difficult for new developers to understand. Existing tools focus on individual aspects such as code quality, security scanning, documentation generation, or deployment automation, forcing developers to switch between multiple platforms.

This fragmented workflow leads to:

- Long onboarding time for new developers.
- Difficulty understanding project architecture.
- Poor code quality due to overlooked issues.
- Security vulnerabilities remaining undetected.

- Time-consuming manual DevOps configuration.
- Lack of centralized project insights.

Therefore, there is a need for a unified AI-powered platform that can analyze repositories, generate intelligent documentation, automate DevOps processes, identify security issues, and provide interactive assistance to developers from a single interface.

3. Objectives

The primary objectives of DevSight AI are:

- Develop an AI-powered repository analysis platform.
- Automatically detect programming languages, frameworks, dependencies, and project architecture.
- Generate interactive architecture and dependency visualizations.
- Analyze code quality and calculate maintainability metrics.
- Detect security vulnerabilities, hardcoded credentials, and outdated libraries.
- Automatically generate deployment files including Dockerfiles, GitHub Actions workflows, and Kubernetes manifests.
- Generate project documentation and architecture summaries using AI.
- Develop an AI chatbot capable of answering repository-specific questions.
- Provide dashboards displaying code quality, security status, and deployment readiness.
- Reduce developer onboarding time and improve overall development productivity.

4. Methodology

The proposed system follows the following workflow:

Step 1: Repository Upload

- User uploads a GitHub repository URL or ZIP project.
- Repository is cloned or extracted.

Step 2: Repository Analysis

The backend scans the project to identify:

- Programming languages
- Frameworks
- Libraries
- Dependencies
- Entry points
- Folder structure

Step 3: Architecture Extraction

The system generates:

- Folder hierarchy
- Module relationships
- Dependency graphs
- Application architecture diagrams

Step 4: Code Quality Analysis

Static analysis tools examine the repository for:

- Code smells
- Duplicate code
- High complexity
- Missing documentation

- Maintainability issues

A quality score is generated.

Step 5: Security Analysis

Security scanners detect:

- Hardcoded API keys
- Passwords
- Secret tokens
- Vulnerable dependencies
- OWASP Top 10 vulnerabilities
- Misconfigurations

Step 6: AI Documentation Generation

The AI generates:

- README improvements
- Architecture summaries
- API documentation
- Installation guide
- Module explanations

Step 7: DevOps Automation

Automatically generate:

- Dockerfile
- Docker Compose
- GitHub Actions CI/CD pipeline

- Kubernetes YAML manifests
- Deployment instructions

Step 8: AI Chat Assistant

Users can ask questions such as:

- "How does authentication work?"
- "Where is the login API?"
- "Explain this service."
- "Show payment flow."

The AI answers using repository context.

Step 9: Dashboard & Reports

Display:

- Quality Score
- Security Score
- Dependency Report
- Deployment Readiness
- Architecture Overview
- Recommendations

5. Tools & Technologies Used

Category	Technologies
Frontend	React.js, Tailwind CSS, TypeScript
Backend	Spring Boot, FastAPI
AI	OpenAI API / Ollama + LangChain
Programming Languages	Java, Python, JavaScript
Database	PostgreSQL
Version Control	Git, GitHub
Containerization	Docker
Orchestration	Kubernetes
CI/CD	GitHub Actions
Static Code Analysis	SonarQube
Security Scanning	Trivy, Sengrep
Documentation	Swagger/OpenAPI
Visualization	Mermaid.js, React Flow
Cloud Deployment	AWS / Azure / Render / Railway

8. Expected Outcomes

- Reduced onboarding time for new developers.
- Improved understanding of software architecture.
- Enhanced code quality and maintainability.
- Early detection of security vulnerabilities.
- Automated generation of deployment configurations.

- AI-assisted repository exploration and documentation.
- Faster, more reliable software delivery.

9. Conclusion

DevSight AI provides a unified platform that integrates Artificial Intelligence, DevOps, and DevSecOps to simplify modern software development. By automating repository analysis, code quality assessment, security scanning, documentation generation, and deployment configuration, the platform reduces manual effort and enables developers to work more efficiently. The AI-powered assistant further enhances productivity by offering contextual explanations and repository-specific guidance, making complex projects easier to understand and maintain. Overall, DevSight AI streamlines the software development lifecycle, improves code reliability and security, and supports faster, more effective collaboration across development teams.